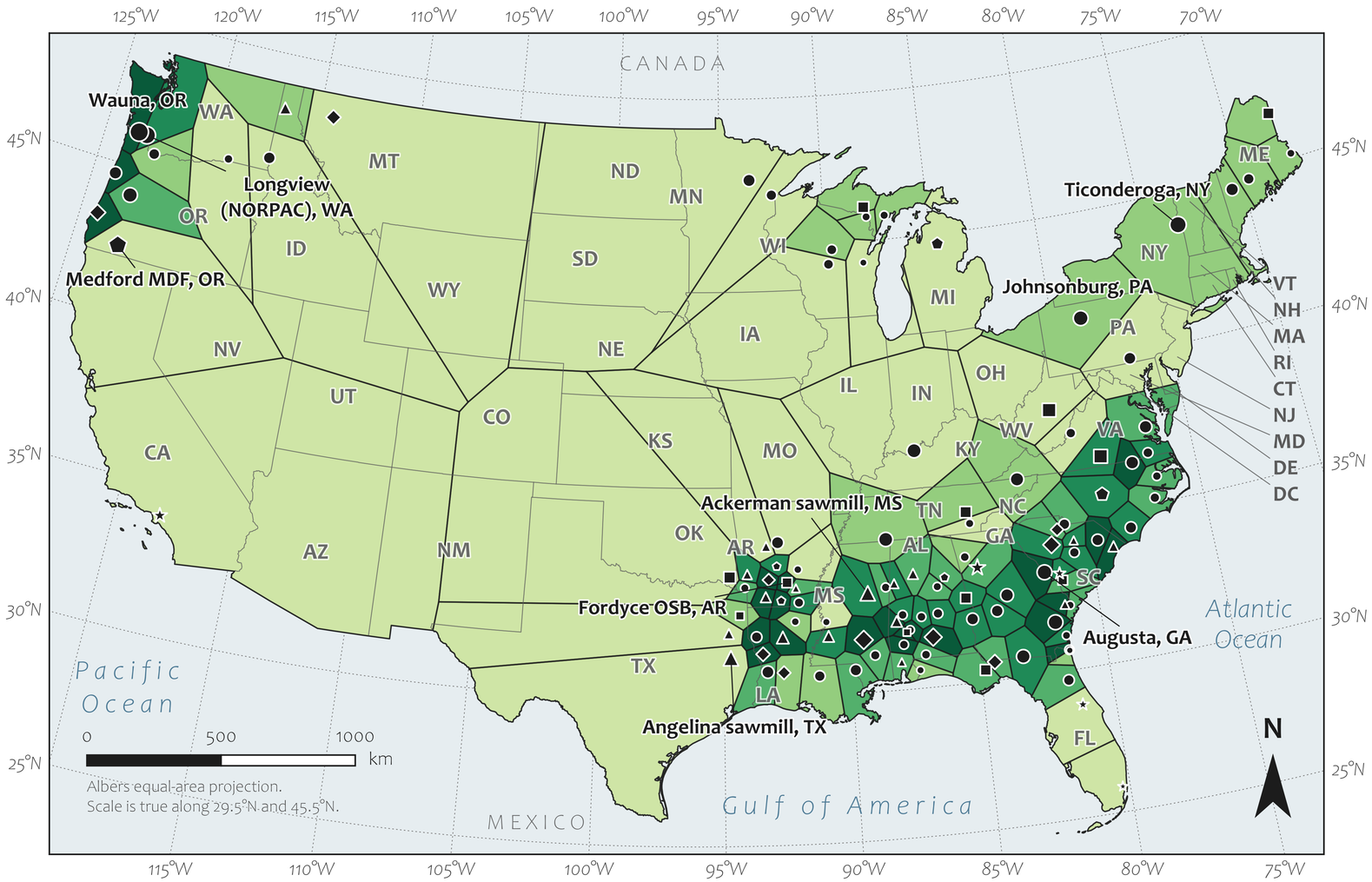
Nearest mill, mapped: Voronoi catchments of U.S. wood mills

Each polygon is the area closer to one mill site than to any other, coloured by the forest wood removed inside it | lower 48 states | a model, not a traffic count



WOOD SUPPLY INSIDE EACH POLYGON

Annual FIA forest removals inside the polygon divided by its area, in thousand ft³ per km² per year. Five quintile classes of 22 to 24 polygons each.

Range	Polygons	Land area	Wood
14.8 to 37.6	23	3%	22%
11.9 to 14.8	24	4%	22%
7.3 to 11.9	23	5%	15%
3.3 to 7.3	22	11%	18%
0.0 to 3.3	24	77%	22%

Land area and wood are shares of the lower 48 total.

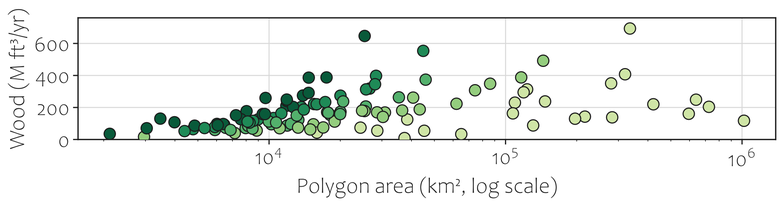
MILL TYPE AND SIZE

- ● Pulp mill (66)
- ▲ Sawmill (17)
- ■ OSB mill (12)
- ◆ Plywood mill (10)
- ⬟ MDF mill (6)
- ★ Paper mill (5)

Symbol size, M ft³/yr: ● 100 ● 300 ● 600

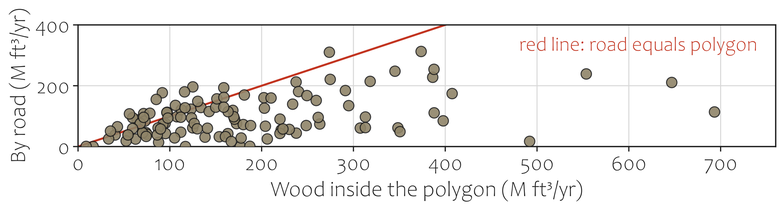
Named on the map: the eight mills with the most wood in their polygon, and the densest polygon.

SIZE OF POLYGON AND WOOD INSIDE



Areas differ 481-fold, wood only 79-fold, so a small polygon mostly means a dense mill network, not a rich forest.

STRAIGHT LINE AGAINST ROAD



92 mills receive less by road than their polygon holds and 24 receive more. Road time within 120 minutes, not straight distance, decides who gets the wood.

HOW TO READ THIS MAP

A Voronoi polygon holds every point that is nearer to its mill than to any other mill, so the map answers one question: which mill is closest in a straight line? Every mill has a polygon. Where mills are few, one polygon covers a whole region; the single California mill owns most of the Southwest. Colour is the wood removed from forests inside the polygon per square kilometre, and symbol size is the same wood as a total. Dark polygons are heavily harvested, and pale ones are lightly harvested or very large.

LIMITATIONS

Trucks follow roads, so a mill's real catchment differs from its polygon; the Day 2 web map shows road time. Wood is cut in forests only, so a polygon's colour is an average over all of its land. The 117 mills are a sample of open mills, so large western polygons show missing mills as much as empty forest. Removals are county totals shared evenly among forest hexes and include land clearing. Two Camas, Washington, mills share one point and therefore one polygon.

Data: USDA Forest Service FIA annual removals by county (hexes of 842 km²); 117 open wood mills in a working sample; Natural Earth states. Polygons built from mill points in spherical Albers and clipped to the lower 48.